SHUFAN SHAHI

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Education

Islamic University of Technology **2022–2026**
B.Sc. in Computer Science and Engineering **CGPA: 3.58**
Coursework: Algorithms, Machine Learning, Deep Learning, Databases, and Networks.

Chuadanga Govt. College **2021**
Higher Secondary Certificate (HSC) **GPA: 5.00/5.00**

Experience

AI Engineer **2026–Present**
GLiTch LABS

- Built RAG architectures, connecting LLMs to vector databases for optimized semantic search.

Student Researcher **2026–Present**
Elite Research Lab

- Support research projects through literature reviews, experiments, and analysis.

Researcher **2025–2026**
IUT Computer Vision Lab

- Researched multimodal AI methods to improve integration and mitigate modality imbalance.

Senior Executive **2025–2026**
Project Aero

- Contributed to drone R&D by applying AI-based object tracking for international UAV competitions.

Research Experience

EthMuSAM : Mutual Scoring With SAM For Zero Shot Anomaly Segmentation **2026**

- Extracted patch-level representations from images using pretrained DINOv2 encoders.
- Localized anomalous regions by comparing test-image embeddings with representations from normal reference images.

Prototype-Guided Adaptive Feature Learning for Multimodal Emotion Recognition in Conversation (MERC) **2026**

- Developed graph-compatible adaptive feature weighting and modality-specific prototype supervision to learn class-discriminative representations before cross-modal reasoning.
- Improved multimodal emotion-recognition performance across multiple benchmark datasets through adaptive feature processing and prototype-guided learning.

Rethinking Visual Features in Graph Based Multimodal Emotion Recognition in Conversations (MERC): From Low Level Pixels to Semantic Representations **2026**

- Developed a semantic visual-feature pipeline that converts utterance-level videos into temporally structured descriptions of facial expressions, gaze, body language, and emotional transitions.
- Encoded the descriptions into dense representations for graph-based multimodal reasoning, improving cross-modal alignment while providing a viable alternative to noisy handcrafted visual features.

Investigating Lightweight Attention Placement in MobileNetV3 for Image Classification on CIFAR-100 **2026**

- Ran a controlled six-variant ablation of ECA attention in MobileNetV3-Small, comparing addition, stacking, and replacement of native SE modules with paired significance testing across seeds.
- Replacing SE with ECA cut parameters by 28.13% (1.61M to 1.16M) with comparable accuracy (70.62% vs. 70.77% baseline).

Deepfake Detection Using Cross-Covariance Image Transformers (XCiT) **2026**

- Trained an XCiT-based classifier to distinguish real from synthetic face images on a DeepfakeBench-derived dataset of over 260K images.
- Achieved 94.4% validation accuracy using linear-complexity cross-covariance attention.

Projects

NeuroEye - Retinal Disease Classification | *PyTorch, EfficientNetV2-L, ResNet-50, ViT*

- Built a weighted ensemble of EfficientNetV2-L, ResNet-50, and ViT-B/16 with learnable fusion weights to classify 10 retinal conditions from fundus images.
- Trained with mixed precision, per-model learning rates, and early stopping; reached 77.9% test accuracy and 0.80 macro F1.

Drone Vision - Aerial Tracking | *PyTorch, YOLOv11X, ByteTrack*

- Fine-tuned YOLOv11X on VisDrone and integrated ByteTrack for human counting and multi-object tracking in aerial footage.

Benhallu - Bengali Hallucination Detection | *Python, RAG, BanglaBERT, scikit-learn*

- Built a retrieval-augmented pipeline to verify Bengali answers against supplied context or a Bengali QA knowledge base.
- Created a hallucination dataset and used fine-tuned BanglaBERT with answer-comparison features to classify responses.

Bengali Speaker Diarization - Speaker Turn Detection | *PyTorch, Silero VAD, ECAPA-TDNN*

- Built a speech pipeline using neural voice activity detection, speaker embeddings, and agglomerative clustering to label timestamped speaker turns.
- Estimated speaker counts with silhouette scores and exported diarization results to CSV.

Technical Skills

Programming Languages: Python, C++, Java
Data and Machine Learning: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas
Databases: MySQL, Firebase, PostgreSQL
Web Development: React, Tailwind CSS, Next.js, Spring Boot
Mobile Development: React Native, Flutter
Visualization: Matplotlib, Seaborn
Development Tools: Git, Linux

Achievements, Awards and Scholarships

- **5th Place, VAND4 International Competition** **2026**
  Zero-Shot Industrial Track; built a zero-shot computer vision pipeline for detecting anomalies in industrial images using foundation models.
- **First Runner-up, IUT Coderush Hackathon** **2025**
  Built an application under tight deadlines in a fast-paced team.
- **7th Place, BAIC Datathon** **2025**
  Analyzed datasets using preprocessing, visualization, and modeling.
- **IUT Partial Scholarship** **2022**
  Secured a partial scholarship covering all four years of B.Sc. studies.
- **Runner-up, State-Level Physics Olympiad** **2018**

Reference

Shahriar Ivan
Assistant Professor, Department of CSE, IUT
Specialization: Machine Learning, Deep Learning, Computer Vision
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